

Spinal reflexes

The written notes for the spinal reflexes walkthrough. Week 5 competencies 1 to 4 and 7.

Step by step: where the signal travels

3 min

The knee jerk, step by step

1. A tap on the patellar tendon suddenly stretches the quadriceps.
2. The muscle spindles in the quadriceps are stretched with it. Mechanically gated channels in their sensory endings open and the endings depolarize: a receptor potential.
3. The spindle sensory neuron fires faster. Its action potentials travel up the femoral nerve toward the spinal cord.
4. The signal enters the cord through the dorsal root at levels L2 to L4. The cell body of this sensory neuron sits in the dorsal root ganglion, outside the cord.
5. Inside the gray matter, the axon branches. One branch synapses directly on an alpha motor neuron to the quadriceps in the ventral horn. This is the only synapse in the arc.
6. The alpha motor neuron fires. Its axon leaves through the ventral root and runs back down the femoral nerve to the quadriceps.
7. At the neuromuscular junction it releases acetylcholine, and the quadriceps contracts. The leg kicks.
8. At the same moment, a second branch excites an inhibitory interneuron. The interneuron releases glycine onto the hamstring motor neuron, which hyperpolarizes, so the hamstrings relax. This is reciprocal inhibition.
9. Other branches carry the information up the cord to the brain, so you become aware of the tap after the kick has already started.

Keeping the spindle working: alpha-gamma coactivation

1. The brain sends a command to contract the quadriceps.
2. Upper motor neurons excite alpha motor neurons, which shorten the working muscle fibers.
3. At the same time they excite gamma motor neurons, whose axons also leave through the ventral root.
4. Gamma motor neurons contract the two ends of the intrafusal fibers inside each spindle.

5. The middle of the spindle stays stretched, so the spindle keeps firing and keeps reporting muscle length as the muscle shortens.

The Golgi tendon reflex, step by step

1. The quadriceps contracts hard and pulls on its tendon.
2. The collagen fibers of the Golgi tendon organ tighten and squeeze its sensory ending.
3. The tendon organ's sensory neuron fires faster as tension rises.
4. The signal enters the cord through the dorsal root. Its cell body is in the dorsal root ganglion.
5. It excites an inhibitory interneuron in the gray matter.
6. The interneuron inhibits the alpha motor neurons of the same muscle.
7. The quadriceps produces less force. This is the classic protective view; the textbook notes the tendon organ mainly informs the CNS about tension.

Withdrawal and crossed extensor, step by step

1. A tack pierces the skin of the right foot. Nociceptors depolarize and fire.
2. The signal travels up the sensory neurons in the sciatic nerve and enters the cord through the dorsal roots. Their cell bodies are in the dorsal root ganglia.
3. In the dorsal horn they synapse on interneurons. The interneurons spread the signal up and down several segments of the cord.
4. Right side: excitatory interneurons fire the motor neurons to the right flexors, which contract and lift the foot.
5. Right side: inhibitory interneurons quiet the motor neurons to the right extensors, which relax.
6. Other interneurons cross the midline of the cord to the left side.
7. Left side: excitatory interneurons fire the motor neurons to the left extensors, which contract and take the body's weight. Inhibitory interneurons quiet the left flexors.
8. The pain signal also travels up the spinothalamic tract to the brain, so you feel the pain after the foot is already lifting.

The reflex arc

1 min

A reflex is an automatic, stereotyped response to a stimulus. Spinal reflexes are integrated in the spinal cord and do not need the brain.

1. Receptor: detects the stimulus (for the knee jerk, the muscle spindle).
2. Sensory (afferent) neuron: carries the signal into the cord through the dorsal root. Its cell body is in the dorsal root ganglion.
3. Integrating center: the spinal cord gray matter, where the sensory neuron synapses.
4. Motor (efferent) neuron: carries the response out of the cord through the ventral root.
5. Effector: the muscle or gland that responds (for the knee jerk, the quadriceps).

Monosynaptic reflex

The sensory neuron synapses directly on the motor neuron: one synapse. Fastest possible reflex. The stretch reflex is the example.

Polysynaptic reflex

One or more interneurons between the sensory and motor neuron. Slower, because each synapse adds a delay of roughly half a millisecond, but able to inhibit, to cross the midline, and to reach several segments.

Cut	Reflex	Sensation from that area	Voluntary movement
Dorsal root	Lost	Lost	Intact
Ventral root	Lost	Intact	Lost

Sorted by	Kinds	Example
Efferent division	Somatic (somatic motor neurons, skeletal muscle) or autonomic (autonomic neurons; smooth and cardiac muscle, glands, adipose tissue)	Knee jerk; pupillary light reflex
Where integrated	Spinal (spinal cord; the brain can modulate but is not required) or cranial (brain)	Knee jerk; pupillary light reflex
When it develops	Innate (genetically determined) or learned, also called conditioned (acquired through experience)	Rooting reflex in a newborn; Pavlov's dogs salivating at a bell
Number of neurons	Monosynaptic (two neurons, one synapse; only somatic motor reflexes) or polysynaptic (one or more interneurons; all autonomic reflexes)	Knee jerk; withdrawal reflex

Kinds of reflexes

1 min

Every neural reflex: stimulus, sensory receptor, sensory afferent neuron, CNS integrating center, efferent neuron, target muscle or gland. Many use negative feedback; some have a feedforward component that starts the response in anticipation, like bracing for a collision.

- Polysynaptic pathways allow divergence (one stimulus, many targets), convergence (many inputs modify one response), and both excitation and inhibition.
- The pupillary light reflex is autonomic, cranial, innate and polysynaptic.
- Autonomic (visceral) reflexes: urination and defecation are spinal reflexes modulated by the brain (toilet training is learned control); the hypothalamus, thalamus and brain stem integrate heart rate, blood pressure, breathing, eating, water balance and body temperature; the brain stem integrates salivating, vomiting, sneezing, coughing, swallowing and gagging.

An autonomic reflex, step by step

1. A stimulus activates a sensory receptor.
2. The sensory neuron carries the signal into the CNS.
3. It synapses on an interneuron in the integrating center.

- The interneuron synapses on a preganglionic autonomic neuron.
- The preganglionic neuron synapses on a postganglionic neuron in an autonomic ganglion, outside the CNS.
- The postganglionic neuron acts on the target: smooth muscle, cardiac muscle, a gland or adipose tissue.

Skeletal muscle reflexes and proprioceptors

1 min

- Proprioceptors in muscles, joint capsules and ligaments sense length, tension and position. Their sensory neurons have cell bodies in the dorsal root ganglia.
 - The CNS integrates the input through excitatory and inhibitory interneurons, mostly subconsciously; some reaches the cortex, and some reflexes can be modulated consciously.
 - Alpha motor neurons carry the output.
 - Extrafusal muscle fibers, the ordinary contractile fibers, are the effectors.
- No neuron inhibits a skeletal muscle directly. Excitation always means contraction; relaxation is the absence of excitation, arranged by inhibition inside the CNS.
 - The muscle spindle and Golgi tendon organ are the main proprioceptors; both are nerve endings in connective tissue capsules with mechanically gated cation channels. Joint receptors contribute less than once thought.
 - Loss of proprioception, with disease or aging, can cause instability and falls.

- Stretch reflex: a protective reflex against overstretching. Stretch speeds spindle firing; the afferent synapses directly on alpha motor neurons of the same muscle; the muscle contracts and the stretch is undone (negative feedback). Example: a book added to an outstretched hand.
- Reciprocal inhibition: spindle afferent branches excite inhibitory interneurons that silence the antagonist's motor neurons. The pathways controlling one joint form a myotatic unit.
- Alpha-gamma coactivation: gamma motor neurons contract the ends of the intrafusal fibers as the alpha motor neurons contract the muscle, keeping the spindle center stretched so it keeps firing.

Alpha motor neuron

Motor neuron to the extrafusal fibers.

Gamma motor neuron

Motor neuron to the contractile ends of the intrafusal fibers.

The Golgi tendon organ

1 min

- Free nerve endings wound between collagen fibers in a capsule at the muscle-tendon junction, in series with the muscle fibers.
- Responds mainly to tension, especially while the muscle pulls without shortening; relatively insensitive to stretch.
- Classic Golgi tendon reflex (autogenic inhibition): tendon organ afferent excites an inhibitory interneuron that inhibits the same muscle's motor neurons.
- Your textbook's update: the tendon organ's main role is sending tension information to CNS integrating centers, combined with spindle input for fine control of posture and movement.

The stretch reflex

1 min

- Muscle spindles lie parallel to the extrafusal fibers; every skeletal muscle has many except one muscle in the jaw. Each spindle holds intrafusal fibers whose ends contract but whose central region lacks myofibrils and is wrapped by sensory endings.
- Spindles are tonically active, slowly adapting stretch receptors: at resting length the center is already stretched enough to fire them.
- Tonic spindle input keeps alpha motor neurons firing, so even a resting muscle has muscle tone, its resistance to stretch.

	Muscle spindle	Golgi tendon organ
Position	In parallel with the extrafusal fibers	In series, at the muscle-tendon junction
Reports	Length (stretch)	Tension (force)
Classic reflex effect on its own muscle	Excites: contract	Inhibits: relax (autogenic inhibition)
Synapses	Monosynaptic to its own muscle	Polysynaptic, through an inhibitory interneuron
Held still while the load rises	Steady firing	Firing rises

Picking up a heavy weight: alpha motor neurons, gamma motor neurons, spindle afferents and tendon organ afferents are all active.

Withdrawal and crossed extensor reflexes 1 min

1. Nociceptors in the right foot fire (stepping on a tack).
 2. The primary sensory neuron enters the cord and diverges onto many interneurons.
 3. Same side: excitatory interneurons drive the alpha motor neurons to the flexors, and inhibitory interneurons silence those to the extensors. The foot lifts: the flexion (withdrawal) reflex.
 4. Other side: interneurons cross the midline; the left extensors contract and the left flexors are inhibited, so the left leg takes the weight: the crossed extensor reflex, a postural reflex.
 5. One collateral ascends to the brain for the sensation of pain and for postural adjustment.
- Both are polysynaptic, so the withdrawal reflex is slower than the monosynaptic stretch reflex.
 - Divergence lets one stimulus control two antagonist pairs on both sides and inform the brain.

Tetanus (lockjaw)

1. Clostridium tetani enters through a wound and releases the toxin tetanospasmin.
2. The toxin is taken up into somatic motor neuron terminals by endocytosis.
3. It travels backward along the axon to the cell body in the spinal cord.
4. It is released and blocks neurotransmitter release at inhibitory synapses (inhibitory transmitters such as glycine and GABA hyperpolarize their targets).
5. With no inhibition, agonists and antagonists contract together: rigid spasms beginning in the jaw and spreading, sometimes to the breathing muscles.
6. Treatment can include antitoxin, antibiotics, muscle relaxants, and a drug that blocks acetylcholine receptors with ventilator support. Immunization prevents it.

The brain's control of spinal reflexes 1 min

- Muscle tone: spindles fire at rest, keeping a few alpha motor neurons active, so a resting muscle is

- slightly firm. Cutting the dorsal roots removes tone and the tendon reflex.
- Descending pathways from the brain act on spinal motor neurons, interneurons and gamma motor neurons, and on balance they hold spinal reflexes in check.
 - Jendrassik maneuver: pulling the hooked hands apart during the tap is thought to raise descending drive and spindle sensitivity, so a weak reflex gets bigger. A reflex that reinforces has an intact arc.
 - Plantar response: stroking the sole normally curls the toes down. A big toe that extends with the toes fanning is the Babinski sign, an upper motor neuron sign. It is normal in infants until about one to two years old, while the corticospinal tract matures.

	Upper motor neuron lesion	Lower motor neuron lesion
Where	Motor pathway in the brain or cord above the reflex (corticospinal tract)	Ventral horn motor neuron, ventral root or peripheral nerve
Strength	Weak	Weak
Tone	High (spasticity)	Low (flaccid)
Reflexes	Brisk, often with clonus	Reduced or absent
Plantar response	Babinski sign	Normal or absent
Over weeks	Little wasting	Muscle wasting and fasciculations

Testing reflexes 1 min

Grade	Meaning
0	Absent
1+	Less than usual
2+	Normal
3+	Brisker than usual
4+	Very brisk, often with clonus

- A normal reflex needs normal conduction in every neuron, normal transmission at the neuromuscular junction, and normal muscle contraction.
- An absent reflex means the arc is broken somewhere: receptor, sensory neuron or dorsal root, cord segment, motor neuron or ventral root, neuromuscular junction, or muscle.

- Not all abnormal reflexes are neuromuscular: slowed relaxation of the ankle reflex suggests hypothyroidism.
- A very brisk reflex with clonus usually means the descending pathways from the brain that normally hold reflexes in check are damaged (an upper motor neuron lesion).
- Compare left with right. A difference between sides matters more than one number.
- Latency separates a reflex from a voluntary response. The knee jerk appears a few tens of milliseconds after the tap. Catching a dropped ruler takes about 150 to 250 milliseconds, because the path runs through the eyes, the brain and down the cord, crossing many more synapses.